

Stakeholders & Data Needs

This document describes the main stakeholders for the Lite dataset, the decisions they care about, and which Bronze tables support those needs.

Scope of the Lite Dataset

The version focuses on a realistic retail supply chain slice across 2026–2028: store + online sales, inventory positions, and inventory movements. The goal is to produce a clean Silver layer that can later support a simple Kimball-style model in Power BI.

Bronze Tables Included

- `bronze__product_master__lite` — product reference data (SKU, name, category, brand, `unit_cost`).
- `bronze__location_master__lite` — location reference data loaded from Excel (stores, DC, web).
- `bronze__sales_transactions__lite` — transactional sales (date/time as raw string, qty, `unit_price`).
- `bronze__inventory_snapshots__lite` — daily on-hand inventory by location and SKU (date as raw string).
- `bronze__inventory_movements__lite` — inventory event stream (sale/replenish/adjustment) with from/to locations.

Stakeholders

- Supply Chain Manager — accountable for availability, service levels, and inventory efficiency.
- Inventory Planner — plans replenishment and monitors stock positions across channels.
- Retail Operations Lead — cares about store execution and repeated stock issues in specific locations.
- E-commerce Lead — monitors online demand, replenishment performance, and availability for WEB channel.
- Finance / Commercial Analyst — interested in revenue impact and inventory value proxy.

Primary Business Questions

1. Availability: Where and when do we run low or out of stock (by SKU and location)?
2. Sales impact: Which SKUs/locations drive most sales volume and revenue proxy (qty × `unit_price`)?
3. Inventory efficiency: Which SKUs are consistently overstocked vs. fast-moving?
4. Replenishment behavior: How often are replenishments happening and for which SKUs/locations?

5. Data alignment: Do movements and snapshots broadly reconcile with sales patterns (directionally)?

Data Needs by Question

1) Availability (low stock / potential stockouts)

Tables used:

- bronze__inventory_snapshots__lite (on_hand_qty by date/location/sku)
- bronze__location_master__lite (location_type, region)
- bronze__product_master__lite (category/brand grouping)

What we need from the data:

- Daily inventory by SKU and location
- Ability to group by store vs web vs DC
- Product grouping to see category patterns

2) Sales impact (volume and revenue proxy)

Tables used:

- bronze__sales_transactions__lite (qty, unit_price; date standardized in Silver)
- bronze__product_master__lite
- bronze__location_master__lite

What we need from the data:

- Daily/weekly sales aggregation
- SKU/category performance
- Store vs web comparison

3) Inventory efficiency (over/understock patterns)

Tables used:

- bronze__inventory_snapshots__lite
- bronze__sales_transactions__lite
- bronze__product_master__lite

What we need from the data:

- Compare average on-hand vs demand proxies
- Find slow movers with persistent high stock

4) Replenishment behavior (frequency and magnitude)

Tables used:

- bronze__inventory_movements__lite (movement_type, qty, from/to)
- bronze__inventory_snapshots__lite

What we need from the data:

- Count and summarize REPLENISH events
- Compare replenishment cadence to stock positions

5) Basic reconciliation / sanity checks

Tables used:

- bronze__inventory_movements__lite
- bronze__sales_transactions__lite
- bronze__inventory_snapshots__lite

What we need from the data:

- Directional consistency (sales reduce inventory; replenishments increase it)
- Identify abnormal negatives or extreme quantities